

summary

Project: Sentiment Analysis of Product Reviews
Input: "This phone is amazing! Battery life is excellent."
Output: Positive / Negative

1. Data Acquisition

What is it?
Collecting text data from various sources.
Sources
• CSV files
• Databases
• Websites (Web Scraping)
• APIs
• Social Media
• User Reviews
Example

Review	Label
This phone is amazing	Positive
Waste of money	Negative

2. Text Preparation

What is it?
Organizing and structuring the collected text.

Tasks
• Create dataset
• Assign labels
• Remove duplicates
• Handle missing values
•
Example
Before:
Review: None
Review: This phone is amazing
Review: This phone is amazing

After:
Review: This phone is amazing

Goal
Prepare clean, usable data.

3. Text Cleanup

What is it?
Removing unwanted elements.

Example
Before:
This phone is AMAZING!!! 🍷🍷
Visit <https://amazon.com>

After:
This phone is amazing

Tasks
• Remove URLs
• Remove HTML tags
• Remove emojis
• Remove punctuation
• Remove special symbols
•
import re

text = re.sub(r"http\S+", "", text)

4. Basic Preprocessing

What is it?
Converting text into a more standardized form.

Tasks
Lowercasing

I LOVE NLP

↓
i love nlp

Tokenization
i love nlp

↓
['i', 'love', 'nlp']

Stopword Removal
['i', 'love', 'nlp']

↓
['love', 'nlp']

Stemming
playing
played
plays

↓
play

Lemmatization
better

↓
good

5. Advanced Preprocessing

What is it?
More sophisticated NLP processing.

Techniques
Part-of-Speech Tagging
I love NLP

↓

I → Pronoun
love → Verb
NLP → Noun

Named Entity Recognition (NER)
Elon Musk lives in Texas

↓

Elon Musk → Person
Texas → Location

Dependency Parsing

Understanding relationships between words.
John eats apples

↓

John → subject
eats → verb
apples → object

6. Feature Engineering

What is it?

Converting text into numbers.
Machine Learning models cannot understand text.

They understand:
1, 2, 3, 4

not
I love NLP

Bag of Words

I love NLP

↓

[1, 1, 1]

TF-IDF

Measures word importance.
excellent

gets higher weight than
the

Word Embeddings

Convert words into dense vectors.

Example:

King
Queen
Man
Woman

become:
[0.3, 0.8, 0.5]

etc.

Examples:

- Word2Vec
- GloVe
- FastText

7. Modelling

What is it?

Choosing an algorithm.

Examples:

- Traditional ML
- Logistic Regression
- Naive Bayes
- SVM
- Random Forest
- Deep Learning
- RNN
- LSTM
- GRU

Transformers

- BERT
- RoBERTa
- DistilBERT
- BanglaBERT

Example:

```
from sklearn.linear_model import LogisticRegression
```

8. Model Building (Training)

What is it?

Teaching the model using training data.

```
model.fit(X_train, y_train)
```

Example:

Input:

This phone is amazing

Output:

Positive

The model learns patterns.

9. Evaluation

What is it?

Checking model performance.

Metrics

Classification:

- Accuracy
- Precision
- Recall
- F1 Score

Example:

accuracy = 92%

Confusion Matrix:
Actual vs Predicted

10. Deployment

What is it?

Making the model available to users.

Example:

Train model →
Save model →
Create API

model.predict("This phone is awesome")

User enters review and gets prediction.

11. Monitoring

What is it?

Watching the model after deployment.

Questions:

- Is accuracy dropping?
- Is data changing?
- Is latency increasing?

Example:

Model trained on 2024 reviews.

Now users use new slang:

This phone is fire 🔥

The model may struggle.

12. Model Update (Retraining)

What is it?

Retraining the model with newer data.

Example:

Old dataset:

2024 reviews

New dataset:

2024 + 2025 + 2026 reviews

Retrain:

model.fit(new_data)

Deploy new version.

Complete NLP Pipeline

Data Acquisition

↓

Text Preparation

↓

Text Cleanup

↓

Basic Preprocessing

↓

Advanced Preprocessing

↓

Feature Engineering

↓

Modelling

↓

Model Building (Training)

↓

Evaluation

↓

Deployment

↓

Monitoring

↓

Model Update

A useful teaching distinction:

- **Text Cleanup + Basic + Advanced Preprocessing** → Make text cleaner and more informative.
- **Feature Engineering** → Convert text into numbers.
- **Modelling + Training** → Learn patterns from those numbers.
- **Evaluation + Deployment + Monitoring** → Put the model into production and keep it working over time.

